Retention Time Prediction

Bonus Challenge – Winning Solution

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challenge_rt_silja_ohlsen.pdf

Winning Hybrid Two-Stage Model Architecture

This two-stage hybrid model successfully predicted retention times for 73,782 peptides on various mass spectrometry instruments (`raw_file`, achieving a Mean Absolute Error (MAE) of **2.783 minutes**.

Model Architecture

Stage 1: Feature Engineering (26 Physicochemical Features)

From each peptide sequence, the model extracts 26 chemical properties: 6 basic features and 20 amino acid counts.

Preprocessing: Modifications are removed from sequences.

6 Basic Features:

- **Length**: Number of amino acids
- **GRAVY Score**: Average hydropathy (A=1.8, V=4.2, D=-3.5, etc.) – positive values indicate hydrophobicity (late elution)
- **Net Charge**: Count of positive (K,R,H) minus negative (D,E) residues
- **Hydrophobic Fraction**: Proportion of 8 most hydrophobic residues (A,I,L,M,F,W,V,Y)
- **Basic Fraction**: Proportion of positive residues (K,R,H)
- **Acidic Fraction**: Proportion of negative residues (D,E)

20 Amino Acid Counts: Frequency of each standard amino acid (A,C,D,...,Y).

These 26 features comprehensively encode the biophysical properties of each peptide.

Stage 2: Universal iRT Prediction (ExtraTreesRegressor)

Training data is split 80:20 (random_state=42). An ExtraTreesRegressor with 600 decision trees (max_depth=30, min_samples_leaf=2) learns the nonlinear mapping from 26 features to `indexed_retention_tim` (iRT).

iRT represents instrument-independent normalized retention times.

Stage 3: Instrument-Specific Calibration (LinearRegression)

For each unique `raw_file`, training data is grouped and separate LinearRegression models map predicted iRT to actual `retention_time`, producing instrument-specific formulas:

$$\text{RT} = m \times \text{pred_iRT} + b$$

All formulas are stored in a dictionary enabling $O(1)$ lookup during prediction.

Stage 4: Production Prediction Pipeline

Prediction data undergoes identical feature extraction and iRT prediction via ExtraTrees. Final RT computation:

- **Known instrument**: $\text{RT} = m \times \text{pred_iRT} + b$
- **Unknown instrument (fallback)**: $\text{RT} = \max(\text{pred_iRT}, 0)$

This produces exactly 73,782 rows in the required `peptide_sequence,retention_time` format.

Score:

- The model achieves a MAE of **2.783**.

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